

Wenfan Wei

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TECHNICAL SKILLS

Programming: C++, C, Python, CUDA, x86 Assembly, Java, SQL

Robotics & Systems: ROS 2, SLAM, Isaac Sim, Isaac Lab, Linux

Machine Learning & Computer Vision: PyTorch, TensorFlow, NumPy, pandas

Developer Tools: Git, Docker, MATLAB, AWS

EXPERIENCE

AMR Automation Intern

August 2026 – Present

Wellwit Robotics

Los Angeles, CA

- Tune and validate onboard AMR sensors and perception parameters to improve data quality and reliability for localization and navigation.
- Use ROS 2 and SLAM pipelines to build maps, evaluate localization, reproduce system issues, collect telemetry, and troubleshoot navigation behavior through system-parameter tuning.

Robotics Intern

May 2026 – August 2026

MAXIEYE

Shanghai, China

- Built an NVIDIA Isaac Sim synthetic-data pipeline for robotic tube grasping with domain randomization, exporting RGB, depth, segmentation, and bounding-box annotations.
- Generated and validated over 2,000 robot demonstrations totaling 350,000 synchronized frames with RGB camera data, 7-DOF joint states, and action targets.
- Fine-tuned YOLO11 segmentation and pi0.5 vision-language-action models on simulated and teleoperated data using cloud and physical GPUs, and conducted reinforcement-learning experiments in Isaac Lab for sim-to-real deployment.
- Aligned RGB segmentation masks with depth, back-projected pixels into 3D point clouds, estimated object centers, and generated grasp candidates ranked by inverse-kinematics feasibility and collision clearance.
- Developed a humanoid robot trajectory-processing pipeline using spike removal, Savitzky-Golay and One Euro filtering, with forward-kinematics validation to reduce motion noise while preserving end-effector accuracy.

AI Developer

June 2025 – August 2025

Amazon Web Services Cal Poly SLO DxHub

San Luis Obispo, CA

- Selected for an 80-person internship cohort from more than 1,400 applicants.
- Built a production-ready NumPy and pandas pipeline to clean and normalize more than 5,000 institutional receipt records.
- Developed an AWS-hosted RAG agent and custom interface to automate receipt auditing and flag policy violations; presented results to Amazon engineers and university stakeholders.

PROJECTS

NormaCore ELRobot

- Integrated and calibrated 6-DOF leader-follower robot arms using ST3215 serial-bus servos and servo driver boards; debugged power, USB-serial, bus communication, motor-ID, and wiring issues across the hardware stack.
- Configured robot joint limits and actuator stiffness/damping for simulation and prepared a leader-follower teleoperation pipeline for collecting manipulation demonstrations.

UAV Geospatial Target Fusion

- Built an Isaac Sim synthetic-perception pipeline to generate UAV imagery with randomized camera poses and environmental conditions, exporting datasets for YOLO object-detection training.
- Developing a multi-view localization pipeline that projects 2D detections into 3D world-space rays using camera intrinsics and poses, then fuses observations to estimate target locations.

Libreye Eagle

- Built a privacy-focused security-camera system using Frigate NVR, Docker, and a fine-tuned YOLO11 object-detection model.
- Designed and 3D-printed a thermally managed enclosure and integrated camera modules, power controls, microSD storage, and battery support.
- Configured FFmpeg, RTSP, and Ethernet streaming on a local server for low-latency video processing and local data ownership.

EDUCATION

University of California, Irvine

Irvine, CA

Bachelor of Science, Computer Engineering

Expected July 2028

- **Activities:** Financial Officer, ZOTBotics; Treasurer, UCI Transfers Club

De Anza College

Cupertino, CA

Associate of Science, Computer Engineering | GPA: 3.6

June 2024 – June 2026

- **Coursework:** Data Structures, Computer Architecture and x86 Assembly, Circuits, Differential Equations, Linear Algebra, Physics
- **Activities:** President, Mathematical Investment Trading Society; Treasurer, Developers Guild; Project Lead, Foothill Computer Engineering Club